Investigating the effect of supports on Ni-based catalysts using in-situ characterization for ammonia decomposition

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Hydrogen energy is considered an efficient green energy system that provides a stable solution for achieving a carbon-neutral future. However, hydrogen storage and transport have been viewed as significant challenges. In this regard, ammonia emerged as an excellent hydrogen carrier with a hydrogen content of 17.8 wt. % as well as its ease of transportation at room temperature under moderate pressures [1]. Ammonia decomposition, being an endothermic reaction, can facilitate the breakdown of ammonia molecules into hydrogen and nitrogen on demand, thus providing an efficient and safe method for hydrogen storage and transportation. Various supported metal catalysts have been studied for ammonia decomposition, primarily focusing on noble metal catalysts [2]. In this work, we focus on a systematic study of the Ni-based catalysts on different supports. We investigate the structure-property relationship and the effect of the supports on the catalytic efficiency using various in-situ characterization methods.

Wet impregnation is used for the synthesis of 10 at.% Ni on different supports among which Ni/CeO_2 showed the highest conversion while Ni/TiO_2 showed the lowest. The difference in efficiency results from interactions between the metal catalyst and its support during activation and the reaction. The analysis of catalysts after their use can lead to significant changes in their properties. Therefore, conducting a thorough examination under reaction conditions is crucial for understanding and optimizing catalysts accurately. Ex-situ characterization such as X-ray diffraction (XRD) and scanning transmission electron microscopy are employed to study the properties of the as-synthesized catalysts. In-situ XRD studies revealed the temperature-induced transformations during activation as well as during the ammonia decomposition reaction. In-situ total scattering (PDF) and in-situ EXAFS analysis are used for the understanding of the local atomic structure. The studies showed the inhomogeneity in the dispersion of metallic nickel on the different supports. Electron microscopy

measurements are used to study the size and dispersion of nickel nanoparticles before and after activation. The strong support interaction between Ni and CeO_2 results in improved metal dispersion, which creates a higher number of active sites for the catalytic process. In contrast, the interaction between Ni and TiO_2 leads to the encapsulation of nanoparticles, resulting in a reduced number of active sites. The results prove the importance of systematic in-situ characterization in understanding metal-support interaction.

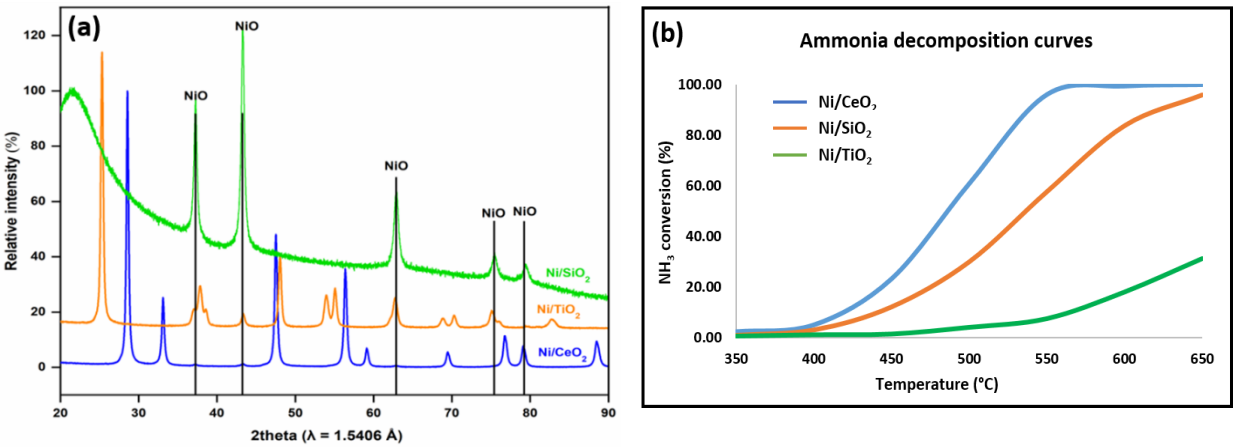

Figure: (a) Ex-situ XRD patterns of Ni-based catalysts on different supports, (b) Ammonia decomposition curves of Ni-based catalysts after activation in 10% H_2/90%N_2 (WHSV = 1500 $cm^3\ g_{Ni}^{-1}\ h^{-1}$).

References:

[1] Kojima, Y. et al. International Journal of Hydrogen Energy, 2022, 47, 22832-22839.

[2] Liu, L. et al. Chemical Reviews, 2018, 118, 4981−5079.

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